COMP 3613 Assignment 2

Testing and REST API report

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1. Project description

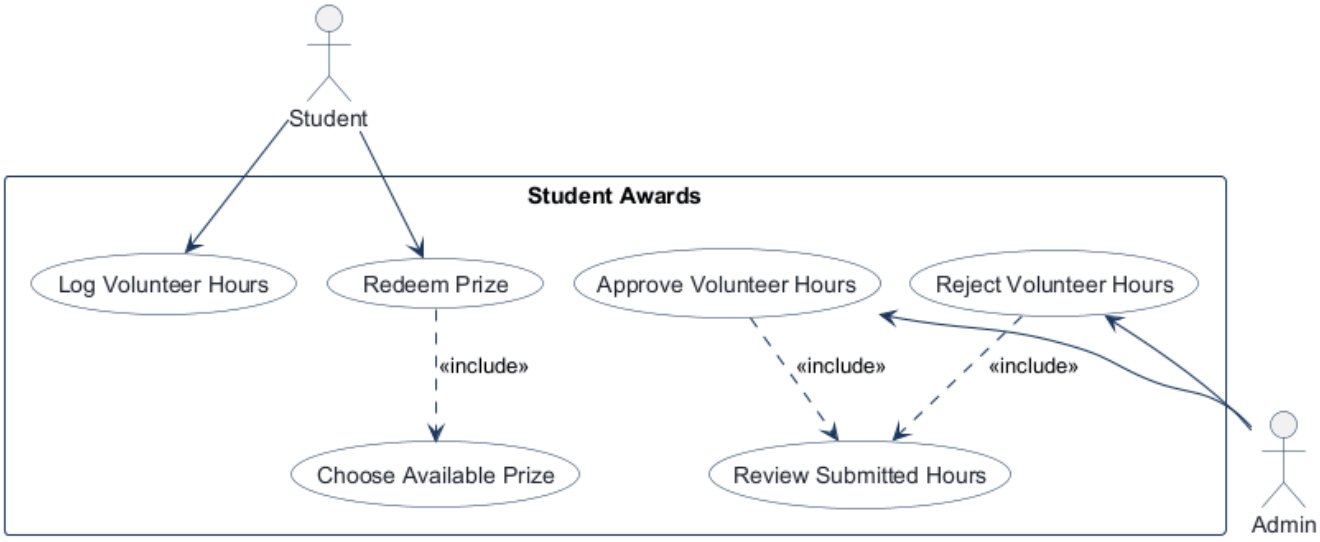
Student Awards (StarServe) is an incentive system for volunteering. Students log the hours they volunteer. An admin reviews each submission and approves or rejects it. A rejection must include a reason. Each approved hour earns one point, and students spend points on prizes from a catalogue. Redeeming a prize deducts its cost from the student's balance and reduces the prize's stock by one.

Main workflows:

1. **Log Volunteer Hours (Student)**: submit date, activity, hours and optional notes. The entry waits for review, and a rejected entry can be resubmitted.
2. **Approve Volunteer Hours (Admin)**: review the pending queue, then approve or reject with a reason.
3. **Redeem Prize (Student)**: browse prizes, confirm the cost against the current balance, and redeem.

Stack: FastAPI, SQLModel, Jinja templates and PostgreSQL on Render (SQLite locally and in tests). Code is layered as Routes -> Services (business rules) -> Repositories (database access) -> Models.

2. Use case diagram



3. Entity relationship diagram

`!Entity relationship diagram`

Points balance = approved hours − sum of points_required for the student's redemptions. It is calculated when needed, not stored.

4. Application wireframes

#	Screen	Wireframe
W1	Student Dashboard	
W2	Log Volunteer Hours	
W3	My Hours	
W4	Available Prizes	
W5	Confirm Redemption	
W6	Admin Dashboard	
W7	Review Queue	
W8	Review Submission	

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5. Test plans

5.1 User Acceptance Test plan

Setup: run python manage.py init and then python manage.py run, and open http://localhost:5000. Accounts: **bob / bobpass** (student, starts with 6 approved hours and 4 pending) and **admin / adminpass** (admin). Use two browsers (or one normal and one private window) so the student and the admin can be logged in at the same time.

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-1	W1	bob	Log in and view the dashboard.	Dashboard shows Welcome bob, Verified Hours 6, Points Balance 6, Pending Submissions 1, the Log Volunteer Hours and Browse Prizes buttons, and a Recent Submissions table with Date, Activity, Hours and Status.	Dashboard showed Welcome bob, Verified Hours 6, Points Balance 6, Pending Submissions 1, both buttons and the Recent Submissions table.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-2	W2	bob	Click + Log Volunteer Hours, enter today's date, activity "Beach cleanup", 3 hours, and a note, then click Submit for Review.	A success message appears. The new entry shows as pending on My Hours, and Pending Submissions on the dashboard goes up by 1.	Success message shown; Beach cleanup (3h) listed as pending on My Hours; Pending Submissions went to 2.	Pass
UAT-3	W2	bob	Open Log Hours, enter 0 hours (or a blank activity) and submit. Then open the form and click Cancel.	The entry is not saved and an error or validation message is shown. Cancel returns to the dashboard without saving.	0 hours was rejected with a validation message and nothing was saved; Cancel returned to the dashboard.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-4	W7, W8	admin	Log in, click Open Review Queue, find bob's "Beach cleanup" entry and click Review.	The queue lists only pending entries with Student, Date, Activity, Hours and Status. The review page shows the student, date, activity, hours and notes, with Approve and Reject buttons and a rejection-reason box.	Queue listed only pending entries with all columns; review page showed the entry details, Approve/ Reject and the reason box.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-5	W8	admin	On that entry, click Reject with the reason box empty.	The entry is not rejected, a "Please provide a rejection reason" message appears, and the entry stays pending.	"Please provide a rejection reason" shown; entry stayed pending.	Pass
UAT-6	W8, W6	admin	Approve the "Beach cleanup" entry, then go to the Dashboard.	You return to the queue with a success message and the entry is gone from the queue. The dashboard's Approved Today count goes up by 1 and Pending Reviews goes down by 1.	Returned to the queue with a success message; entry removed; Approved Today +1 and Pending Reviews -1.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-7	W8, W3	admin, then bob	Admin: reject bob's remaining pending entry with the reason "Hours do not match the sign-in sheet". Bob: open My Hours.	The entry shows rejected with the reason displayed below the table and a Resubmit action. Filter: Rejected shows only rejected entries, and Filter: All shows every entry.	Entry shown as rejected with the reason and a Resubmit link; Rejected filter showed only that entry, All showed every entry.	Pass
UAT-8	W3, W2	bob	Click Resubmit on the rejected entry, change the hours, and submit.	The form opens pre-filled with the rejected entry's details and the rejection reason. After submitting, a new pending entry appears on My Hours.	Form opened pre-filled with the reason shown; a new pending entry appeared after submitting.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-9	W4	bob	Open Prizes.	The balance is shown ("N points available "). Each prize shows name, description, cost and stock. Prizes bob can afford show Redeem prize, and the others show Need more hours.	"9 points available " shown; prizes listed with cost and stock; affordable ones showed Redeem prize, Festival Pass showed Need more hours.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-10	W5	bob	Click Redeem prize on an affordable prize, check the confirmation, then click Cancel.	The confirmation shows the prize, its cost, Balance now and Balance after (now minus cost). Cancel returns to Prizes with no change to balance or stock.	Confirmation showed Cost 4, Balance now 9, Balance after 5; Cancel changed nothing.	Pass
UAT-11	W5, W4, W1	bob	Redeem the same prize again and click Confirm.	A success message appears. The prize now shows Redeemed, its stock is down by 1, and the points balance on Prizes and the Dashboard is down by the prize cost.	Success message; Campus Coffee Card showed Redeemed, stock 5 to 4; balance 5 on Prizes and Dashboard.	Pass

ID	Wireframe	User	Task	Success criteria	Actual result	Pass/Fail
UAT-12	W6	bob	While logged in as bob, go to http://localhost:5000/admin.	Access is refused (401 Unauthorized page) and no admin data is shown.	401 Unauthorized page shown; no admin data visible.	Pass

5.2 Unit test plan

AwardService is tested in isolation, using a FakeAwardRepository in place of the database (tests/test_unit_awards.py).

ID	Unit	Input	Expected
U1	create_log	blank activity	ValueError, nothing saved
U2	create_log	hours = 0	ValueError, nothing saved
U3	create_log	" Beach cleanup ", notes " "	activity trimmed, notes None
U4	decide_log	action "maybe"	ValueError
U5	decide_log	rejected, no notes	ValueError
U6	decide_log	rejected, whitespace notes	ValueError
U7	decide_log	approved, no notes	saved with None notes
U8	redeem_prize	unknown prize	ValueError (NotFoundError)
U9	redeem_prize	hours below cost	ValueError

ID	Unit	Input	Expected
U10	redeem_prize	hours equal to cost	redemption created
U11	redeem_prize	8 approved, 5 already spent, cost 5	ValueError (balance is 3)

5.3 Integration test plan

These run the real app with FastAPI TestClient against a fresh SQLite database for each test. Student and admin fixtures log in through /login (cookie) or /api/login (Bearer token).

Web flows (tests/test_integration_awards.py)

ID	Flow	Expected
I1	Student logs hours	log saved as pending
I2	Admin approves it	status approved, student's approved total increases
I3	Admin rejects without reason	blocked, log stays pending
I4	Admin rejects with reason	status rejected, reason saved
I5	Student redeems affordable prize	redemption recorded, stock down by 1
I6	Student redeems again	refused ("already redeemed")
I7	Student redeems too-expensive prize	refused
I8	Student requests /admin	refused (401)
I9	Anonymous requests /app	refused (401)
I10	Admin dashboard counts	pending, approved today, rejected today are correct

ID	Flow	Expected
I11	Redeeming spends points	second 5-pt prize refused with 8 hours; page shows 3 points
I12	Admin approves a non-existent log	handled with a message, no server error
I21	My Hours filter, reason, resubmit	filter shows only rejected; reason shown; resubmit form pre-filled

REST API (tests/test_api_awards.py)

ID	Flow	Expected
I13	POST /api/login	200 with bearer token and role; wrong password 401
I14	GET /api/logs without token	401 JSON
I15	Student calls admin endpoints	403
I16	POST/GET /api/logs	201 pending with trimmed activity; hours 0 -> 400; list returns own logs
I17	Admin GET/PATCH /api/admin/logs	pending filter; reject without reason 400; bad status 422; missing log 404; approve 200
I18	GET /api/admin/dashboard	correct counts
I19	POST /api/prizes/{id}/redemptions	201; again 409; too expensive 400; unknown 404
I20	Balance, redemptions, prizes	balance 8 -> 3 after a 5-pt prize; redemption listed; stock down by 1

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6. Implementation

Repository: https://github.com/JohnathanLaldeo/comp3613a1

6.1 REST API

All endpoints are under /api, return JSON and require Authorization: Bearer <token> except login.

Method	Path	Who	Purpose	Success	Errors
POST	/api/login	anyone	Get a token	200	401
GET	/api/logs	student	List my volunteer logs	200	401
POST	/api/logs	student	Log hours	201	400, 401, 422
GET	/api/me/balance	student	Approved hours, points spent, points	200	401
GET	/api/prizes	student	List active prizes	200	401
POST	/api/prizes/{id}/redemptions	student	Redeem a prize	201	400 (not enough points), 404, 409 (already redeemed / out of stock)
GET	/api/redemptions	student	List my redemptions	200	401
GET	/api/admin/logs?status=	admin	List logs, optional status filter	200	401, 403, 422

Method	Path	Who	Purpose	Success	Errors
PATCH	/api/admin/logs/{id}	admin	Approve or reject ({"status", "admin_notes"})	200	400, 401, 403, 404, 422
GET	/api/admin/dashboard	admin	Pending / approved today / rejected today	200	401, 403

Authorization: a missing or invalid token returns **401**. A logged-in student calling an admin endpoint returns **403**. Student endpoints only ever return the caller's own data. The web pages enforce the same rules through the login cookie.

REST conventions: URLs name resources rather than actions (/logs, /prizes/{id}/redemptions). GET reads, POST creates and returns 201, and PATCH updates part of a log (its status). Errors return a JSON {"detail": ...} with a matching status code (400, 401, 403, 404, 409 or 422).

6.2 Tests

Suite	File	Command
Unit	tests/test_unit_awards.py	python manage.py test unit
Integration (web + API)	tests/test_integration_awards.py, tests/test_api_awards.py	python manage.py test integration
Everything	tests/	python manage.py test or pytest
API (Postman)	postman/student-awards.postman_collection.json	Postman Collection Runner, or npx newman run postman/student-awards.postman_collection.json

The Postman collection has 28 requests in 4 folders (auth and authorization, student logs, admin review, prizes and points), and each request has test scripts. It logs in as both users, stores their tokens, and follows the full workflow: log hours, then reject without a reason (400), then approve, then check hours increased, then redeem, then redeem again (409), then check points were spent, then try a prize that costs too much (400). Run python

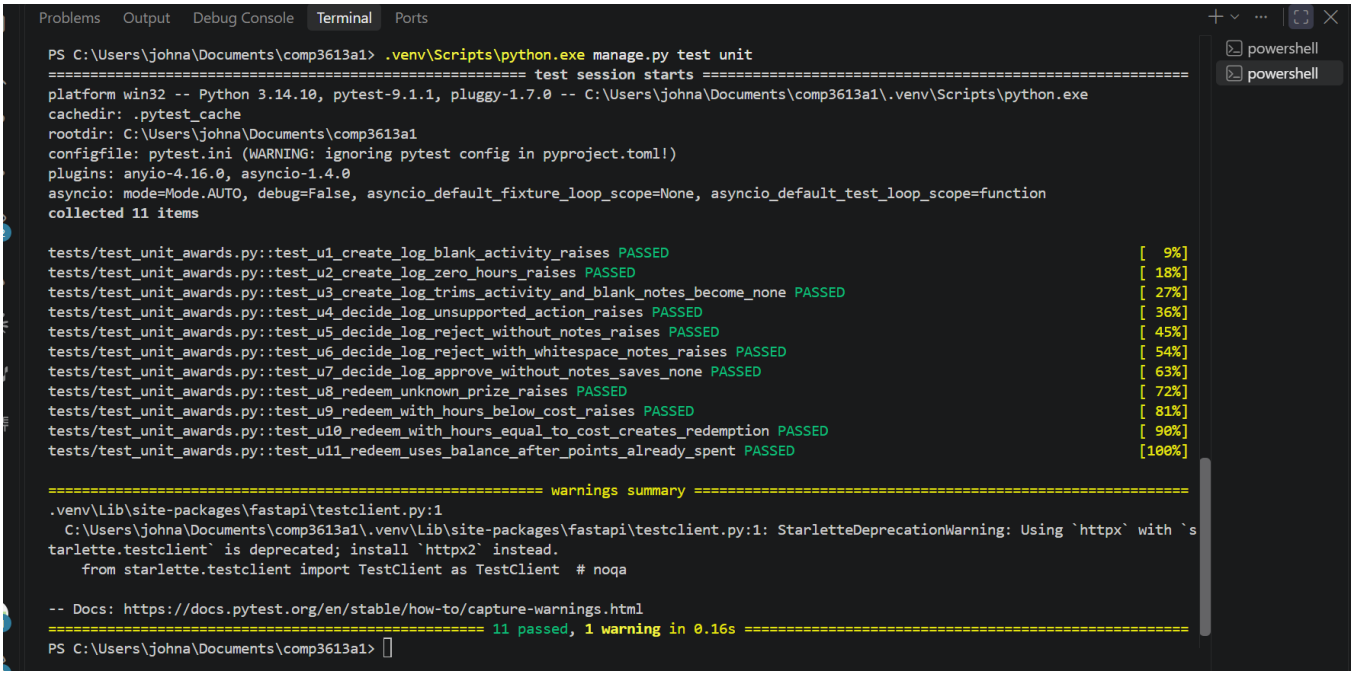
manage.py init before each collection run so it starts from the seed data.

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7. Reports

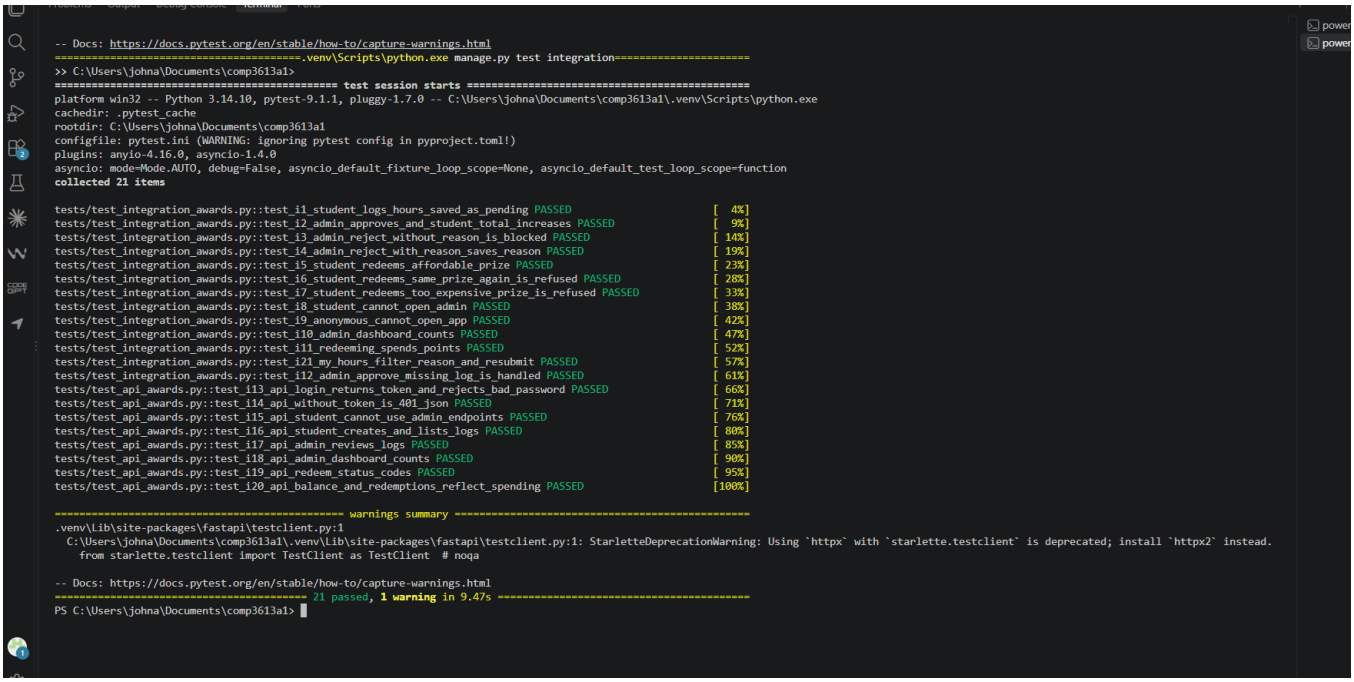
7.1 Unit tests report

Screenshot of python manage.py test unit (11 passed):



7.2 Integration tests report

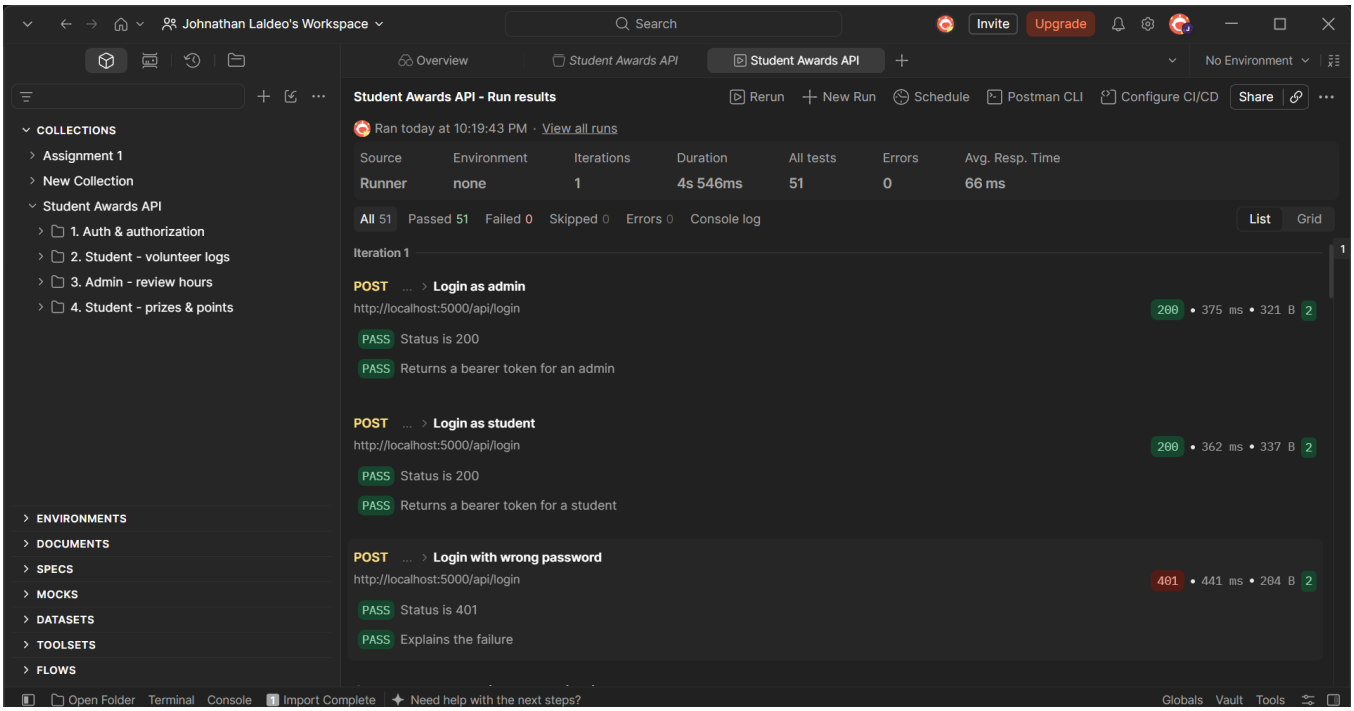
Screenshot of python manage.py test integration (21 passed):



7.3 Postman collection run report

_Collection postman/student-awards.postman_collection.json imported into the Postman app and run with the Collection Runner against a freshly seeded app (python manage.py init). Result: 51 tests, 51 passed, 0 failed, 0 errors. The same collection was also run from the

command line with Newman (npx newman run postman/student-awards.postman_collection.json); its machine-readable report is in postman/newman-run-report.json._



7.4 User Acceptance Test report

UAT was carried out by the developer in the browser against a freshly seeded local app (python manage.py init), with bob in one window and admin in a private window. Actual results are recorded in section 5.1.

Total UAT cases	Passed	Failed	Notes
12	12	0	All acceptance criteria met; no defects found.